

Title

Psychosocial responses and engagement intentions following prostate cancer polygenic risk disclosure: secondary qualitative findings and design recommendations for clinical implementation

Abstract

Beyond cognitive comprehension, patient-centered risk communication must account for psychological and behavioral responses and decision-making intentions. Secondary analysis of qualitative interviews examined emotional reactions to prostate-cancer polygenic-risk disclosures, intentions to pursue screening based on disclosed risk level, and participants' perceived autonomy in clinical decision-making. Participants reported primarily neutral-to-positive emotional responses to risk disclosure; notably, no participant expressed anxiety, distress, or negative psychological impact from learning their polygenic risk status. Hypothetical risk-disclosure context (moderate versus high polygenic-risk scenarios) meaningfully influenced perceived screening relevance and stated screening intentions. Participants with higher-risk scenarios endorsed stronger intentions to pursue screening. Most participants explicitly valued having autonomous decision-making authority and declined preference for directive clinical guidance. Supplementary design features--interactive risk calculators, hyperlinks to specialist consultations, downloadable summary documents--were valued as enhancing usability and facilitating secondary conversations with family or primary-care clinicians. Several participants noted that portable or shareable document formats would support productive clinical discussions. These secondary insights suggest that thoughtfully-designed risk-communication materials can convey polygenic information while preserving patient autonomy and minimizing psychological burden, supporting responsible clinical implementation.